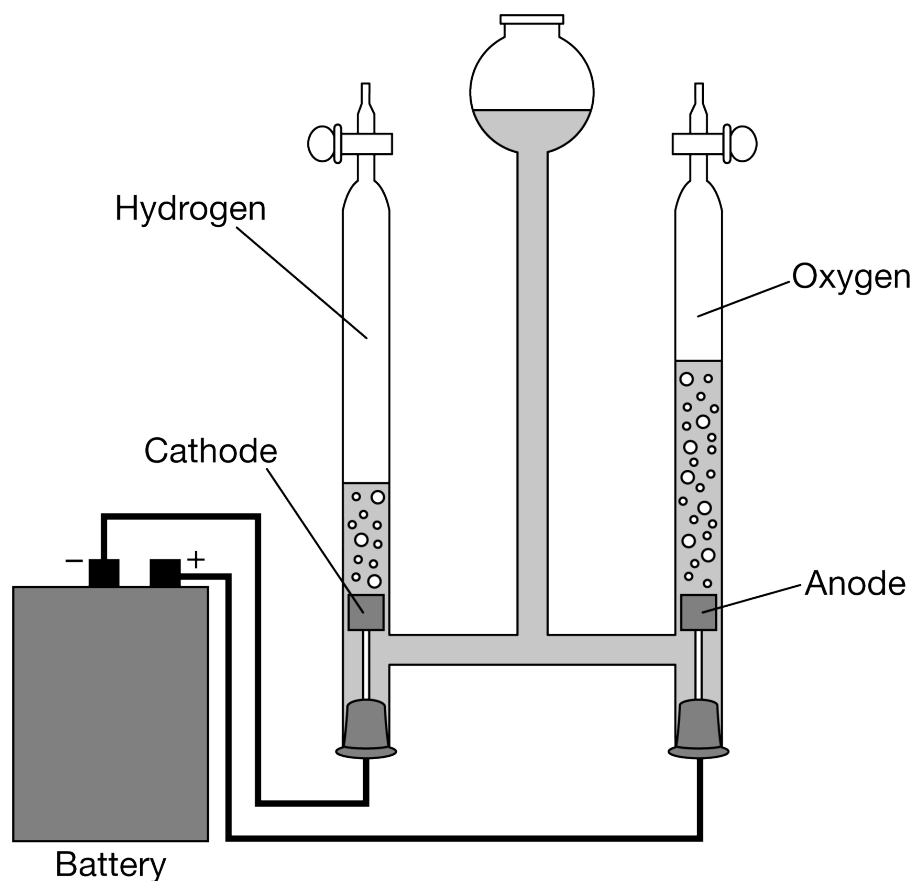


Chapter 12 Part 2: Electrolysis

1.

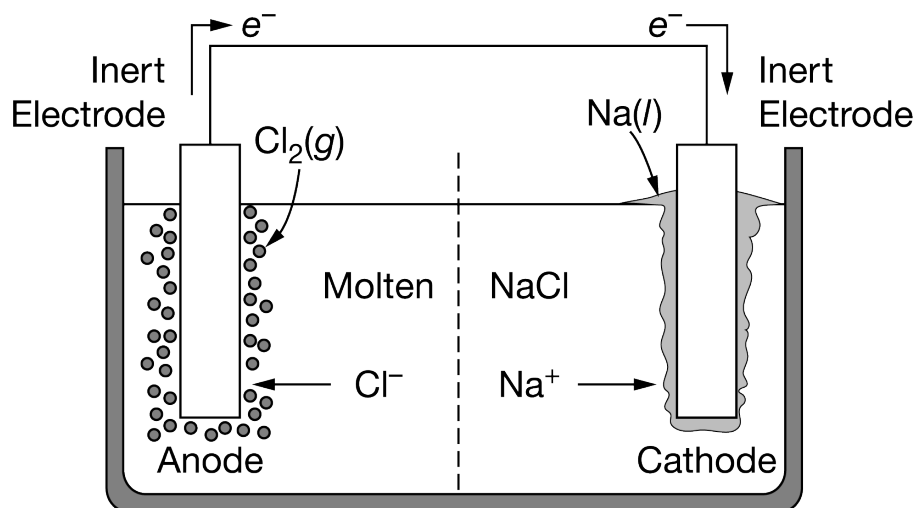


The reaction in which $\text{H}_2\text{O}(l)$ is decomposed into $\text{H}_2(g)$ and $\text{O}_2(g)$ is thermodynamically unfavorable ($\Delta G^\circ > 0$). However, an electrolytic cell, such as the one represented above, can be used to make the reaction occur. Which of the following identifies a flaw in the representation?

- (A) Oxidation is occurring at the anode instead of at the cathode.
- (B) The equation for the reaction is not correctly balanced. ✓
- (C) Electrical energy is needed for the reaction to proceed.
- (D) The relative volumes of collected gas shown in each tube are incorrect.

Chapter 12 Part 2: Electrolysis

2.



The decomposition of $\text{NaCl}(l)$ into $\text{Na}(l)$ and $\text{Cl}_2(g)$ is thermodynamically unfavorable. The decomposition requires the input of energy from an external source. The diagram represents an electrolytic cell that can be used to drive the decomposition reaction. Which of the following identifies a flaw in the representation?

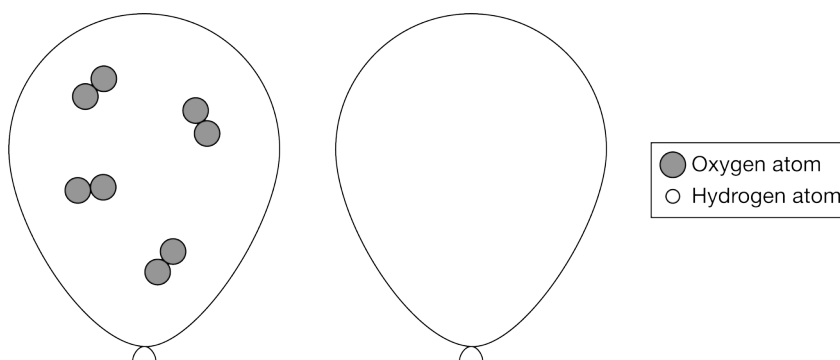
- (A) Oxidation is occurring at the anode.
- (B) Molten Na is shown at the cathode.
- (C) An external source of energy is not shown. ✓
- (D) The direction of the electron flow in the wires is incorrect.

Chapter 12 Part 2: Electrolysis

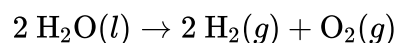
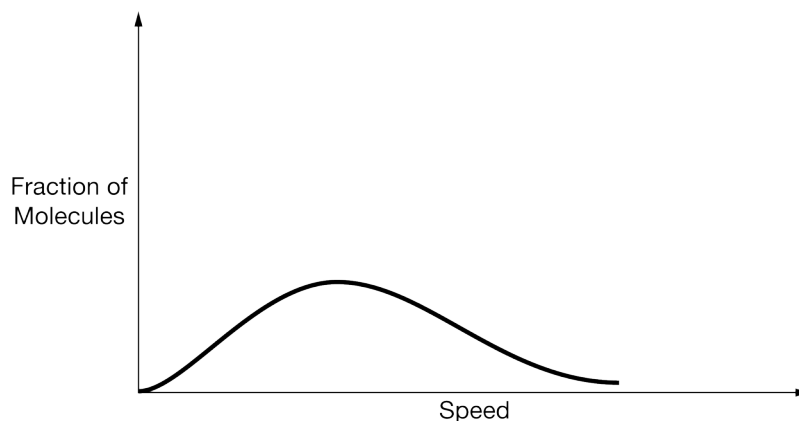
3. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about hydrogen and oxygen gases.

- (a) A particle-level representation of $\text{O}_2(g)$ is shown in the balloon below on the left. The balloon on the right contains the same volume of $\text{H}_2(g)$ at the same temperature and pressure. In the balloon on the right, draw a representation of $\text{H}_2(g)$ with the appropriate number and distribution of particles.

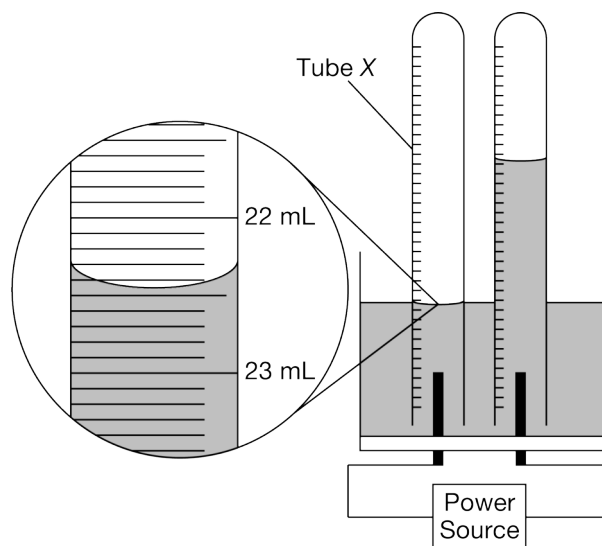


- (b) The graph below shows the Maxwell-Boltzmann distribution for $\text{H}_2(g)$. On the same graph, draw the distribution curve for $\text{O}_2(g)$ at the same temperature.

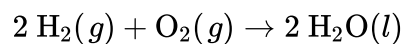


To generate the $\text{H}_2(g)$ and the $\text{O}_2(g)$, a student performs electrolysis of water, represented by the equation above. The student collects the gases in tubes and then adjusts the position of one of them so that the water levels inside and outside of the tube are the same, as shown in the following diagram.

Chapter 12 Part 2: Electrolysis



- (c) Is the gas collected in tube X $\text{H}_2(g)$ or $\text{O}_2(g)$? Justify your answer.
- (d) What volume should the student report for the gas in tube X ?
- (e) Is $\text{H}_2(g)$ produced at the anode or cathode? Justify your answer in terms of oxidation numbers.
- (f) The atmospheric pressure in the lab was 0.989 atm at 22.0°C . The vapor pressure of water at 22.0°C is 0.026 atm. Calculate the pressure of dry gas in tube X .
- (g) The value of E° for the electrolysis of water is -1.23 V . In a second trial of the experiment, the power source was replaced with a voltmeter. Would the volume of hydrogen gas produced be greater than, less than, or equal to the volume produced in the original experiment? Justify your answer.
- (h) The student considers the reverse reaction and proposes a mechanism with a single, elementary step as shown below.



Explain, in terms of particle collisions, why this mechanism is unlikely to be correct.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



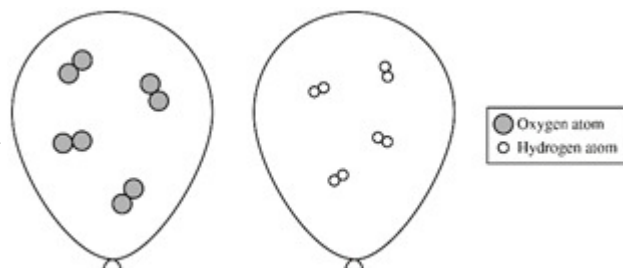
0	1	2
---	---	---

The response meets **all** of the following criteria.

☐

The response shows exactly eight hydrogen atoms as four diatomic molecules.

Chapter 12 Part 2: Electrolysis



- ☐ The response shows particles that are evenly distributed.
· See example given.

Part B

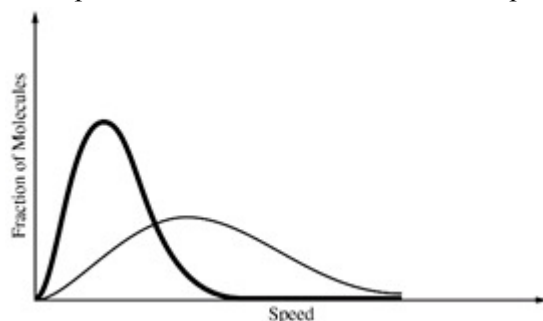
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **all** the following criteria.

- ☐ The response shows a curve that has a taller peak than H_2 .



- ☐ The response shows a curve that has a maximum to the left of the peak for H_2 and approaches zero BEFORE the H_2 curve.
· See example given.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

· Hydrogen. The stoichiometry of the reaction indicates that there should be twice as many moles of hydrogen produced as oxygen, and the volume of gas is proportional to the number of moles.

Chapter 12 Part 2: Electrolysis**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer.

· 22.45 mL

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

· Cathode. Hydrogen is reduced from +1 to 0 in the electrolysis process, and reduction occurs at the cathode.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

· $P_{gas} = 0.989 \text{ atm} - 0.026 \text{ atm} = 0.963 \text{ atm}$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

Chapter 12 Part 2: Electrolysis

· Less than. The reaction is thermodynamically unfavorable (negative E_{cell}°), so the battery is needed to drive the reaction. Therefore, no hydrogen would be produced without the battery.

OR

· Less than. A voltmeter only measures electrical potential and does not supply it.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



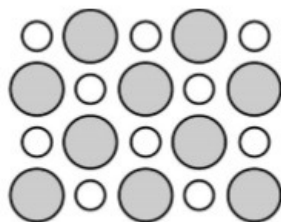
0	1
---	---

The response provides the correct answer and a valid justification.

· Three-body (termolecular) elementary reactions are unlikely because of the low probability of three molecules colliding simultaneously.

Chapter 12 Part 2: Electrolysis

4.

 H_2O molecule

LiCl crystal

The structures of a water molecule and a crystal of $\text{LiCl}(s)$ are represented above. A student prepares a 1.0 M solution by dissolving 4.2 g of $\text{LiCl}(s)$ in enough water to make 100 mL of solution.

- a. In the space provided below, show the interactions of the components of $\text{LiCl}(aq)$ by making a drawing that represents the different particles present in the solution. Base the particles in your drawing on the particles shown in the representations above. Include only one formula unit of LiCl and no more than ten molecules of water. Your drawing must include the following details.

- identity of ions (symbol and charge)
- the arrangement and proper orientation of the particles in the solution

 $\text{LiCl}(aq)$

- b. The student passes a direct current through the solution and observes that chlorine gas is produced at the anode. Identify the chemical species produced at the cathode and justify your answer using the information given in the table below.

Chapter 12 Part 2: Electrolysis

Half-reaction	Standard Reduction Potential at 25°C (V)
$\text{Li}^+(aq) + e^- \rightarrow \text{Li}(s)$	– 3.05
$2 \text{H}_2\text{O}(l) + 2 e^- \rightarrow \text{H}_2(g) + 2 \text{OH}^-(aq)$	– 0.83

Part A

1 point is earned for a correctly drawn and labeled particulate representation of the ions. (Representation must indicate that the smaller ion is Li^+ . Representations that include more than one formula unit of LiCl (dissolved or undissolved) are acceptable as long as at least one of them is separated, labeled, and charged.)

1 point is earned for a correctly drawn particulate representation of water molecules of hydration surrounding the ions.

1 point is earned for correctly representing the orientation of the water molecules of hydration with the proper polarity.

The sketch should clearly show:

1. a clear representation of at least one Li^+ ion and one Cl^- ion separated from each other, labeled, and charged;
2. each ion surrounded by at least two H_2O molecules; and
3. H_2O molecules with the proper orientation around each ion (i.e., the oxygen end of the water molecules closer to the lithium ion and the hydrogen end of the water molecules closer to the chloride ion).



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for a correctly drawn and labeled particulate representation of the ions. (Representation must indicate that the smaller ion is Li^+ . Representations that include more than one formula unit of LiCl (dissolved or undissolved) are acceptable as long as at least one of them is separated, labeled, and charged.)

1 point is earned for a correctly drawn particulate representation of water molecules of hydration surrounding the ions.

1 point is earned for correctly representing the orientation of the water molecules of hydration with the proper polarity.

The sketch should clearly show:

1. a clear representation of at least one Li^+ ion and one Cl^- ion separated from each other, labeled, and charged;
2. each ion surrounded by at least two H_2O molecules; and
3. H_2O molecules with the proper orientation around each ion (i.e., the oxygen end of the water molecules closer to the lithium ion and the hydrogen end of the water molecules closer to the chloride ion).

Part B

Chapter 12 Part 2: Electrolysis

1 point is earned for correctly identifying either of the chemical species produced at the cathode with the proper justification.

Note: The justification must clearly indicate that “higher” means “less negative.” A “lower magnitude” negative value also earns the point.

$\text{H}_2(\text{g})$ and $\text{OH}^-(\text{aq})$

The hydrogen atoms in H_2O are reduced to H_2 at the cathode because this reaction has a higher (more favorable or less negative) standard reduction potential than the reduction of lithium ions to $\text{Li}(\text{s})$.



0

1

The student response earns one of the following points:

1 point is earned for correctly identifying either of the chemical species produced at the cathode with the proper justification.

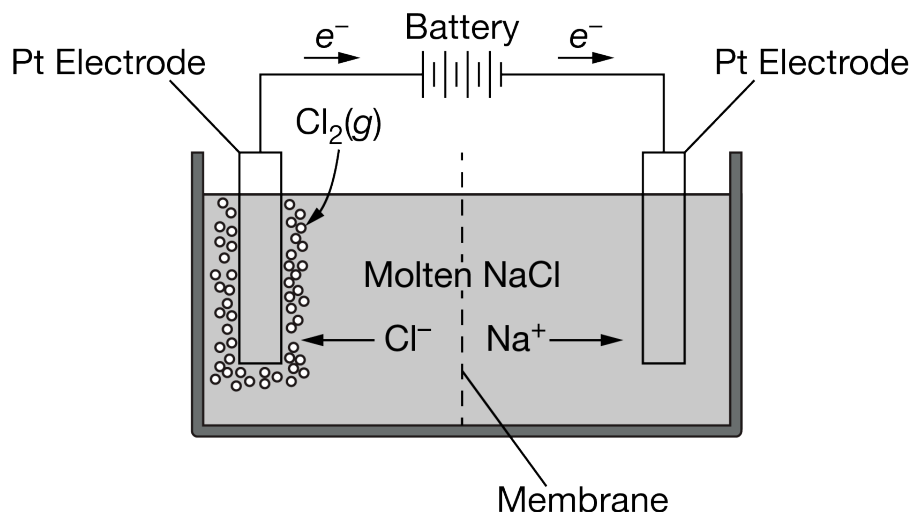
Note: The justification must clearly indicate that “higher” means “less negative.” A “lower magnitude” negative value also earns the point.

$\text{H}_2(\text{g})$ and $\text{OH}^-(\text{aq})$

The hydrogen atoms in H_2O are reduced to H_2 at the cathode because this reaction has a higher (more favorable or less negative) standard reduction potential than the reduction of lithium ions to $\text{Li}(\text{s})$.

Chapter 12 Part 2: Electrolysis

5.

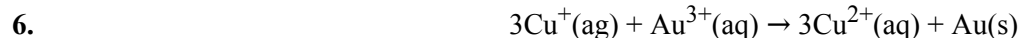


Half-Reaction	Standard Reduction Potential (V)
$\text{Na}^+ + e^- \rightarrow \text{Na}(l)$	-2.71
$\text{Cl}_2(g) + 2e^- \rightarrow 2\text{Cl}^-$	$+1.36$

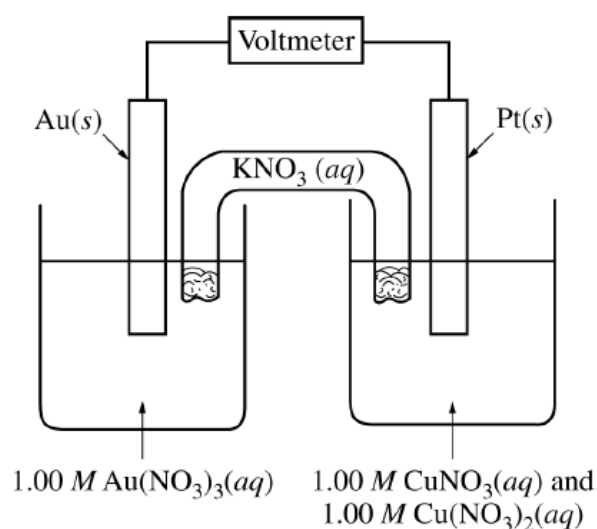
The diagram above represents an electrolytic cell in which the reaction $2\text{NaCl}(l) \rightarrow 2\text{Na}(l) + \text{Cl}_2(g)$ takes place. The table gives the relevant reduction half-reactions and the standard reduction potentials. Based on the information given, which of the following is true?

- (A) The operation of the cell generates a potential of 1.35 V because the reaction is thermodynamically favorable.
- (B) The operation of the cell generates a potential of 5.43 V because the reaction is thermodynamically favorable.
- (C) The operation of the cell requires at least 4.07 V to be supplied because the reaction is not thermodynamically favorable. ✓
- (D) The operation of the cell requires at least 2.72 V to be supplied because the reaction is not thermodynamically favorable.

Chapter 12 Part 2: Electrolysis



A student studying the reaction represented above constructs a voltaic cell as shown in the diagram below. The cell contains an Au(s) electrode in a beaker containing 1.00 M Au(NO₃)₃(aq) and a Pt(s) electrode in a beaker containing 1.00 M CuNO₃(aq) and 1.00 M Cu(NO₃)₂(aq).



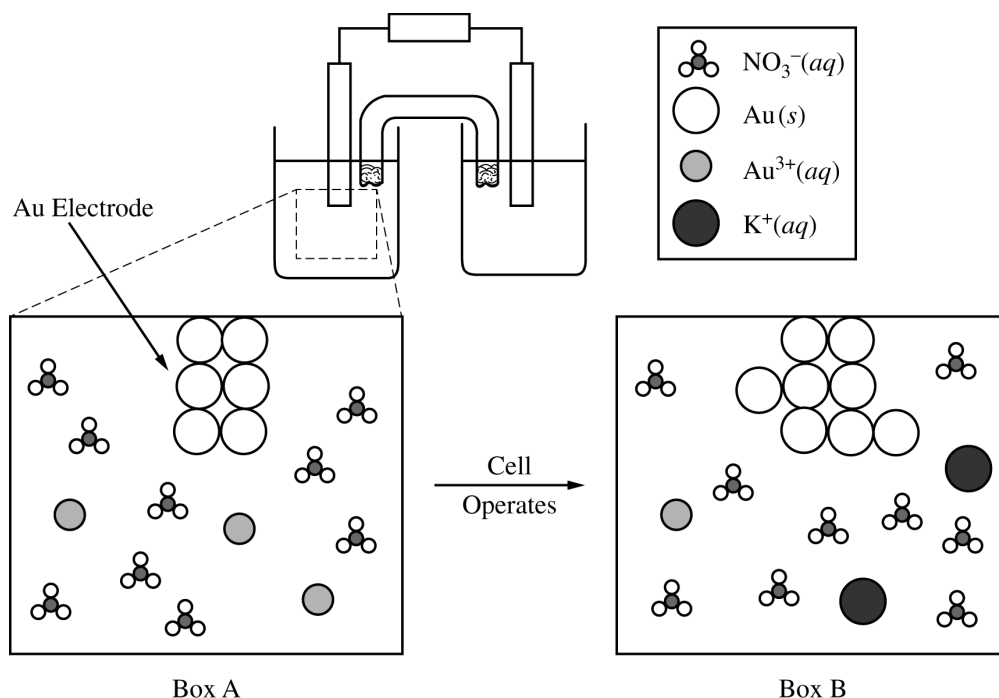
The standard reduction potentials associated with the cell are given in the following table.

Half-reaction	E° (V)
$\text{Cu}^{2+}(\text{aq}) + e^- \rightarrow \text{Cu}^+(\text{aq})$	0.16
$\text{Au}^{3+}(\text{aq}) + 3e^- \rightarrow \text{Au}(\text{s})$	1.50

- Calculate the value of the standard cell potential, E° .
- Does the mass of the Pt(s) electrode increase, decrease, or remain the same as the cell operates? Justify your answer.

An expanded view of the Au half-cell before the reaction occurs is shown in box A below. An expanded view of the Au electrode after the cell operates for a period of time is shown in box B below.

Chapter 12 Part 2: Electrolysis



- c. Assuming that the representation in box A is accurate, explain what is wrong with the stoichiometry represented in box B.
- d. After the cell has operated for 30.0 minutes, 0.185 g of $\text{Au}(s)$ is deposited on the Au electrode.
- Calculate the number of moles of $\text{Au}(s)$ deposited.
 - Calculate the average current that passed through the cell during the deposition of $\text{Au}(s)$.

The student adds some $\text{NaCl}(s)$ to the half-cell that contains $\text{Cu}^+(aq)$ and $\text{Cu}^{2+}(aq)$, and a precipitate forms.

- e. Based on the information in the following table, determine the identity of the precipitate. Justify your answer with a calculation.

Compound	Solubility (mol/L)	K_{sp}
CuCl	?	1.7×10^{-7}
CuCl_2	5.2	?

- f. Write the net ionic equation for the formation of the precipitate.
- g. The student predicts that the cell voltage will increase after the precipitate forms. Do you agree with the

Chapter 12 Part 2: Electrolysis

student's prediction? Justify your answer.

Part A

1 point is earned for the correct answer.

$$E^{\circ} = 1.50 \text{ V} - 0.16 \text{ V} = 1.34 \text{ V}$$



0	1
---	---

Student earns 1/1 point(s)

1 point is earned for the correct answer.

$$E^{\circ} = 1.50 \text{ V} - 0.16 \text{ V} = 1.34 \text{ V}$$

Part B

1 point is earned for the correct answer and a valid justification.

The mass of the Pt(s) electrode remains the same because the Pt does not react and no Cu atoms will be deposited on the Pt electrode.



0	1
---	---

Student earns 1/1 point(s)

1 point is earned for the correct answer and a valid justification.

The mass of the Pt(s) electrode remains the same because the Pt does not react and no Cu atoms will be deposited on the Pt electrode.

Part C

1 point is earned for a valid explanation.

In a redox reaction, the number of electrons lost must equal the number gained, so the net charge must be zero. The total charge of the cations is less than the total charge of the anions.



0	1
---	---

Chapter 12 Part 2: Electrolysis

Student earns 1/1 point(s)

1 point is earned for a valid explanation.

In a redox reaction, the number of electrons lost must equal the number gained, so the net charge must be zero.
The total charge of the cations is less than the total charge of the anions.

Part D

1 point is earned for the correct calculation of the number of moles.

Part i

$$0.185\text{gAu} \times \frac{1\text{molAu}}{196.97\text{gAu}} = 9.39 \times 10^{-4}\text{molAu}$$

Part ii

1 point is earned for the correct number of electrons transferred (may be implicit).

1 point is earned for the correct calculation of the current.

$$I = \frac{q}{t} = \frac{9.39 \times 10^{-4}\text{molAu} \times \frac{3\text{mole}^-}{1\text{molAu}} \times 96,485 \frac{\text{C}}{\text{mole}^-}}{30.0\text{min} \times \frac{60\text{sec}}{1\text{min}}} \\ = 0.151 \text{ C/s} = 0.151 \text{ A}$$



0	1	2	3
---	---	---	---

Student earns 3/3 point(s)

1 point is earned for the correct calculation of the number of moles.

Part i

$$0.185\text{gAu} \times \frac{1\text{molAu}}{196.97\text{gAu}} = 9.39 \times 10^{-4}\text{molAu}$$

Part ii

1 point is earned for the correct number of electrons transferred (may be implicit).

1 point is earned for the correct calculation of the current.

$$I = \frac{q}{t} = \frac{9.39 \times 10^{-4}\text{molAu} \times \frac{3\text{mole}^-}{1\text{molAu}} \times 96,485 \frac{\text{C}}{\text{mole}^-}}{30.0\text{min} \times \frac{60\text{sec}}{1\text{min}}} \\ = 0.151 \text{ C/s} = 0.151 \text{ A}$$

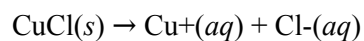
Part E

Chapter 12 Part 2: Electrolysis

1 point is earned for a correct calculation of the solubility of CuCl

1 point is earned for identifying the precipitate with the lower solubility.

For CuCl:



$$K_{sp} = [\text{Cu}^+][\text{Cl}^-]$$

$$1.7 \times 10^{-7} = x^2$$

$$x = 4.1 \times 10^{-4} M = \text{solubility of CuCl}$$

CuCl has a lower solubility compared to CuCl₂ and will precipitate first.



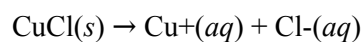
0	1	2
---	---	---

Student earns 2/2 point(s)

1 point is earned for a correct calculation of the solubility of CuCl

1 point is earned for identifying the precipitate with the lower solubility.

For CuCl:



$$K_{sp} = [\text{Cu}^+][\text{Cl}^-]$$

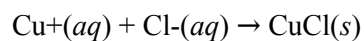
$$1.7 \times 10^{-7} = x^2$$

$$x = 4.1 \times 10^{-4} M = \text{solubility of CuCl}$$

CuCl has a lower solubility compared to CuCl₂ and will precipitate first.

Part F

1 point is earned for the equation consistent with the answer to part (e).

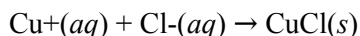


0	1
---	---

Student earns 1/1 point(s)

Chapter 12 Part 2: Electrolysis

1 point is earned for the equation consistent with the answer to part (e).

**Part G**

1 point is earned for a correct choice and a valid justification.

Disagree

The precipitate will decrease $[\text{Cu}^{+}]$, which will cause Q to increase and become closer to the value of K . Therefore, the voltage of the cell will decrease.



0

1

Student earns 1/1 point(s)

1 point is earned for a correct choice and a valid justification.

Disagree

The precipitate will decrease $[\text{Cu}^{+}]$, which will cause Q to increase and become closer to the value of K . Therefore, the voltage of the cell will decrease.

7. An electric current of 1.00 ampere is passed through an aqueous solution of $\text{Ni}(\text{NO}_3)_2$. How long will it take to plate out exactly 1.00 mol of nickel metal, assuming 100 percent current efficiency?

(1 faraday = 96,500 coulombs = 6.02×10^{23} electrons)

(A) 386,000 sec

(B) 193,000 sec

(C) 96,500 sec

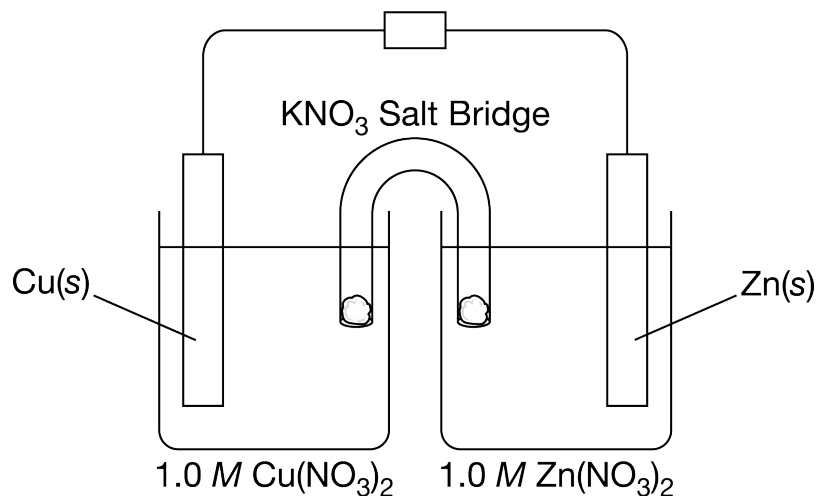
(D) 48,200 sec

(E) 24,100 sec



Chapter 12 Part 2: Electrolysis

8. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

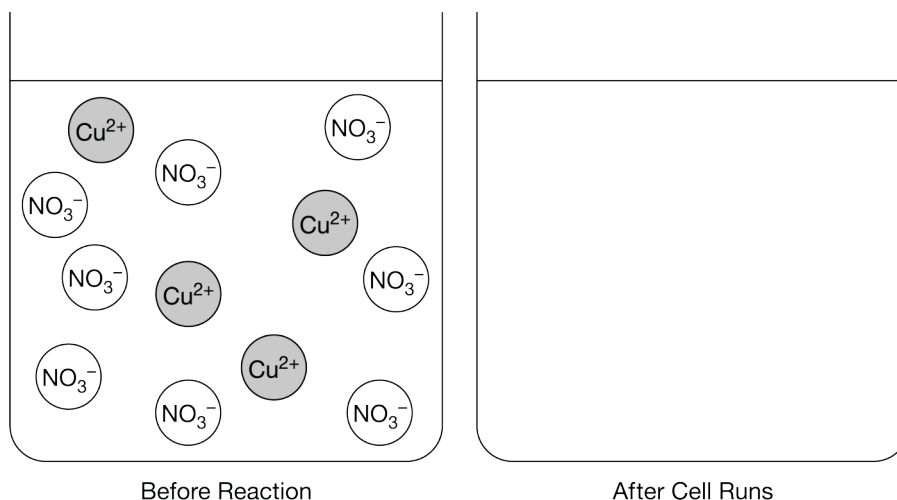


Answer the following questions related to the standard galvanic cell represented in the diagram above. The reduction half-reactions for Cu^{2+} and Zn^{2+} are given in the table below.

Half-Reaction	E° (V)
$\text{Cu}^{2+} + 2 e^- \rightarrow \text{Cu}(s)$	0.34
$\text{Zn}^{2+} + 2 e^- \rightarrow \text{Zn}(s)$	-0.76

- Write the net-ionic equation for the overall reaction that takes place as the cell operates.
- Calculate the value of E° for the cell.
- Calculate ΔG° , in kJ/mol_{rxn} , for the cell.
- The following diagram on the left represents the copper nitrate solution in the half-cell before the galvanic cell starts to operate. In the following diagram on the right, draw a representation of the solution in the same half-cell after the galvanic cell has been operating for some time and is still producing a voltage.

Chapter 12 Part 2: Electrolysis



(e) Assume that the mass of the Cu electrode changes by x grams in a certain period of time. Write a mathematical expression for the change in mass of the Zn electrode during the same period of time.

(f) For the standard cell, the Cu^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Cu}(\text{NO}_3)_2$ and the Zn^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Zn}(\text{NO}_3)_2$. The experiment was repeated, but this time the Cu^{2+} half-cell was made with 0.50 L of 2.0 M $\text{Cu}(\text{NO}_3)_2$ and the Zn^{2+} half-cell was made with 1.0 L of 1.0 M $\text{Zn}(\text{NO}_3)_2$. Is the cell potential for the nonstandard cell greater than, less than, or equal to the value calculated in part (b)? Justify your answer.

(g) If the experiment is repeated with the Cu electrode replaced with a Cu electrode with twice the mass, will the initial cell voltage be greater than, less than, or equal to the initial cell voltage in the original experiment? Justify your answer.

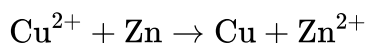
Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives the following equation:

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 2: Electrolysis



0	1
---	---

The response gives the following calculation:

$$E_{cell}^{\circ} = 0.34 \text{ V} - (-0.76 \text{ V}) = 1.10 \text{ V}$$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If the student makes an error, the student can still earn credit for following steps using incorrect values.



0	1	2
---	---	---

The response meets both of the following criteria:

- ☐ The response gives the correct setup.

$$\Delta G^{\circ} = -nFE_{cell}^{\circ} = -(2 \text{ mol})(96,485 \frac{\text{J}}{\text{V mol}})(1.10 \text{ V})$$
- ☐ The response shows the correct result of the calculation.

$$\Delta G^{\circ} = -212,000 \text{ J/mol}_{rxn} = -212 \text{ kJ/mol}_{rxn}$$

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The diagram should be consistent with the reaction written in part (a). If the equation in part (a) is reversed, the diagram should contain five, six, or seven Cu^{2+} ions.



0	1	2
---	---	---

The response meets both of the following criteria:

- ☐ The diagram contains one, two, or three Cu^{2+} ions.
- ☐ There is a charge balance in the diagram (the student can add K^{+} ions from the salt bridge or put in fewer NO_3^{-} ions).

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 2: Electrolysis



0	1
---	---

The response indicates the following:

$$x \text{ g of Cu} \times \frac{1 \text{ mol Cu}}{63.55 \text{ g Cu}} \times \frac{1 \text{ mol Zn}}{1 \text{ mol Cu}} \times \frac{65.38 \text{ g Zn}}{1 \text{ mol Zn}} = \text{g of Zn}$$

$$\text{Change in mass of Zn electrode} = -\frac{65.38}{63.55} x \text{ g}$$

Part (f)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets both of the following criteria:

☐

The response gives the following justification:

$$E = E^\circ - \frac{RT}{nF} \ln Q = E^\circ - \frac{RT}{nF} \ln \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = E^\circ - \frac{RT}{nF} \ln \frac{1}{2}$$

The log of a number less than one is negative, so for the nonstandard cell a negative number is subtracted from the standard potential: If $E = E^\circ - \frac{RT}{nF}$ (negative number), then E will be greater than E° .

☐

The response gives the following conclusion: The cell potential for the nonstandard cell is greater than that for the standard cell (or E is greater than E°).

Part (g)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



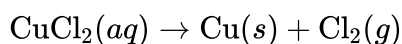
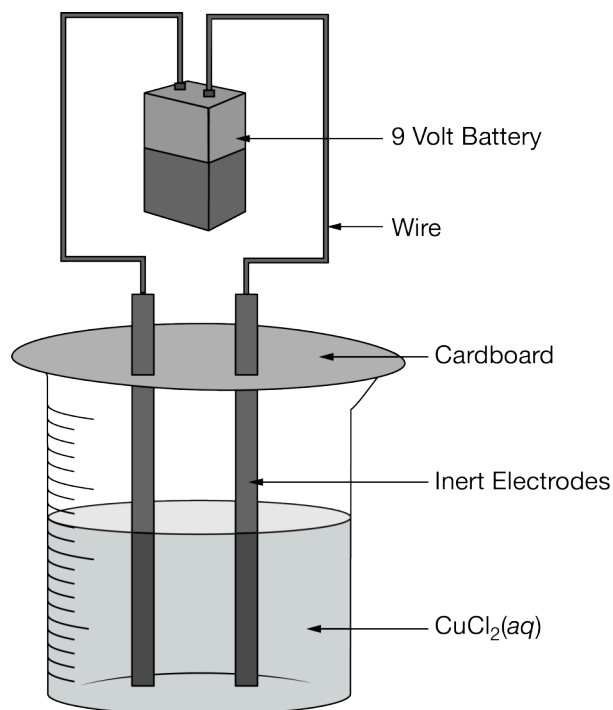
0	1
---	---

The response indicates the following:

The initial voltage when the mass of the Cu electrode is twice the original mass will be equal to the voltage in the original experiment. The mass and concentration of Cu(s) are not in the equation for the cell potential and do not affect it.

Chapter 12 Part 2: Electrolysis

9.



The diagram above shows the apparatus used for the electrolysis of $\text{CuCl}_2(aq)$ according to the chemical equation given. Which of the following mathematical expressions can be used to calculate the current required to produce 0.125 mol of $\text{Cu}(s)$ in 1 hour?

(A) $I = \left(\frac{2 \times 0.125 \times 96,485}{3,600} \right) \text{A}$ ✓

(B) $I = \left(\frac{2 \times 0.125 \times 96,485}{1} \right) \text{A}$

(C) $I = \left(\frac{0.125 \times 96,485}{3,600} \right) \text{A}$

(D) $I = \left(\frac{0.125 \times 96,485}{1} \right) \text{A}$

10. In the electroplating of nickel, 0.200 faraday of electrical charge is passed through a solution of NiSO_4 . What mass of nickel is deposited?

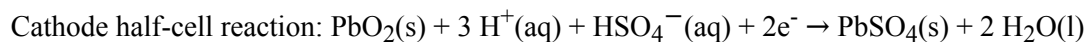
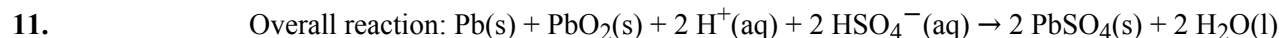
(A) 2.94 g

(B) 5.87 g ✓

(C) 11.7 g

(D) 58.7 g

(E) 294 g

Chapter 12 Part 2: Electrolysis

The equations above represent reactions associated with the operation of a lead storage battery. The first is the overall reaction that occurs as the battery produces an electrical current, and the second is the half-reaction that occurs at the cathode.

- Determine the oxidation number of sulfur in the overall reaction.
- Write the equation for the half-reaction that occurs at the anode as the battery operates.

After the battery has operated for some time, it can be recharged by applying a current to reverse the overall reaction.

- Calculate the time, in seconds, needed to regenerate 100. g of Pb(s) in the battery by applying a current of 5.00 amp

Part A**1 point(s) maximum**

1 point is earned for the correct answer.

+6



0	1
---	---

Student response earns 1 of the following 1 point(s)

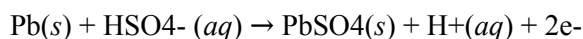
1 point(s) maximum

1 point is earned for the correct answer.

+6

Part B**1 point(s) maximum**

1 point is earned for a correct equation.



Chapter 12 Part 2: Electrolysis

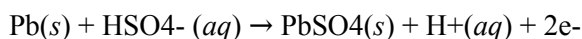


0	1
---	---

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for a correct equation.



Part C

2 point(s) maximum

1 point is earned for the correct number of moles of electrons (may be implicit).

$$100. \text{ g Pb} \times \frac{1 \text{ mol Pb}}{207.2 \text{ g}} = 0.483 \text{ mol Pb}$$

$$0.483 \text{ mol Pb} \times \frac{2 \text{ mol } e^-}{1 \text{ mol Pb}} = 0.966 \text{ mol } e^-$$

$$0.966 \text{ mol } e^- \times 96,485 \text{ C/mol } e^- = 93,200 \text{ C}$$

1 point is earned for the correct time based on the moles of electrons.

$$I = \frac{q}{t} \Rightarrow t = \frac{q}{I}$$

$$\frac{93,200 \text{ C}}{5.00 \text{ amp}} \times \frac{1 \text{ amp}}{1 \text{ C/s}} = 18,600 \text{ s}$$



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the correct number of moles of electrons (may be implicit).

$$100. \text{ g Pb} \times \frac{1 \text{ mol Pb}}{207.2 \text{ g}} = 0.483 \text{ mol Pb}$$

Chapter 12 Part 2: Electrolysis

$$0.483 \text{ mol Pb} \times \frac{2 \text{ mol e}^-}{1 \text{ mol Pb}} = 0.966 \text{ mol e}^-$$

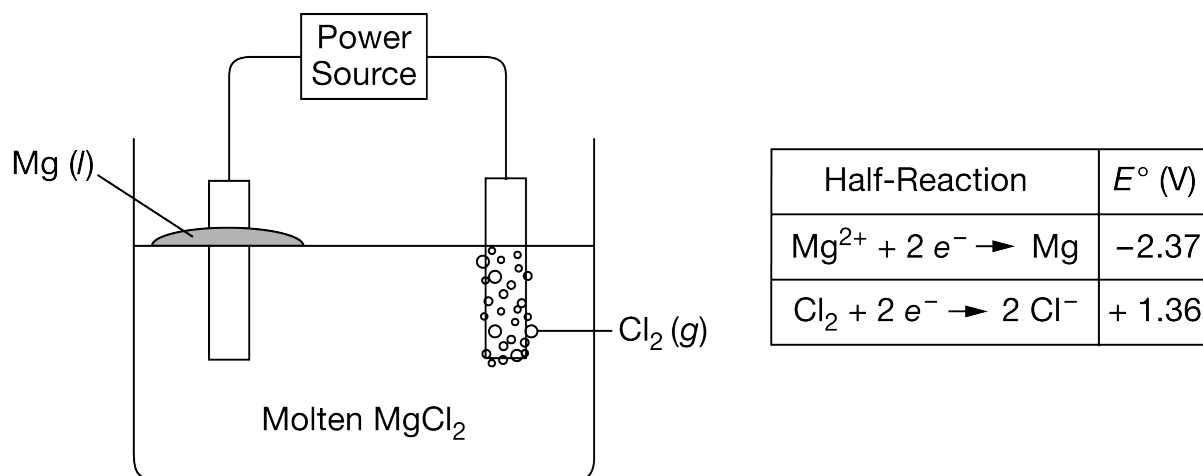
$$0.966 \text{ mol e}^- \times 96,485 \text{ C/mol e}^- = 93,200 \text{ C}$$

1 point is earned for the correct time based on the moles of electrons.

$$I = \frac{q}{t} \Rightarrow t = \frac{q}{I}$$

$$\frac{93,200 \text{ C}}{5.00 \text{ amp}} \times \frac{1 \text{ amp}}{1 \text{ C/s}} = 18,600 \text{ s}$$

12. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Molten MgCl_2 can be decomposed into its elements if a sufficient voltage is applied using inert electrodes. The products of the reaction are liquid Mg (at the cathode) and Cl_2 gas (at the anode). A simplified representation of the cell is shown above. The reduction half-reactions related to the overall reaction in the cell are given in the table.

- (a) Draw an arrow on the diagram to show the direction of electron flow through the external circuit as the cell operates.
- (b) Would an applied voltage of 2.0 V be sufficient for the reaction to occur? Support your claim with a calculation as part of your answer.
- (c) If the current in the cell is kept at a constant 5.00 amps, how many seconds does it take to produce 2.00 g of $\text{Mg}(l)$ at the cathode?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 12 Part 2: Electrolysis

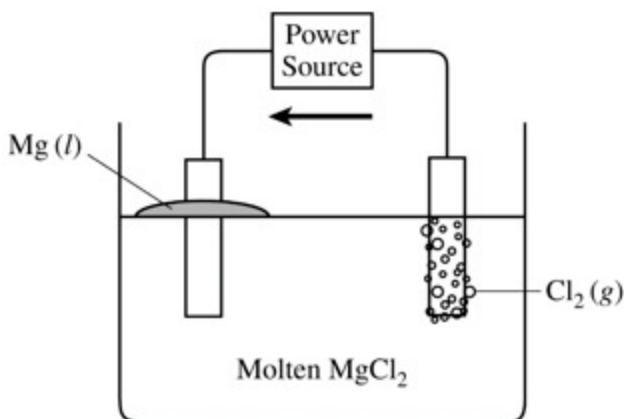
✓

0

1

The response provides the correct answer.

· Electron flow should be indicated only in a counter-clockwise direction in the external circuit, from the Cl_2 anode to the Mg cathode.



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0

1

The response provides the correct answer and a calculated value.

· No, because 2.0 V is less than 3.73 V , which is the minimum voltage needed for electrolysis to occur.

$$E_{\text{cell}}^{\circ} = -2.37\text{ V} + (-1.36\text{ V}) = -3.73\text{ V}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0

1

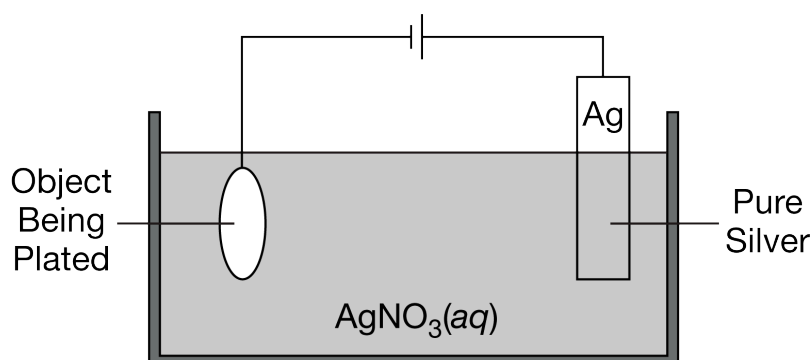
2

The response meets **all** the following criteria.

Chapter 12 Part 2: Electrolysis

- ☐ The response provides the correct calculated value of moles of electrons (may be implicit).
- $$\cdot 2.00 \text{ g Mg} \times \frac{1 \text{ mol Mg}}{24.30 \text{ g Mg}} \times \frac{2 \text{ mol } e^-}{1 \text{ mol Mg}} = 0.165 \text{ mol } e^-$$
- ☐ The response provides the correct calculated number of seconds.
- $$\cdot 0.165 \text{ mol } e^- \times \frac{96,485 \text{ C}}{1 \text{ mol } e^-} \times \frac{1 \text{ s}}{5.00 \text{ C}} = 3180 \text{ s}$$

13.

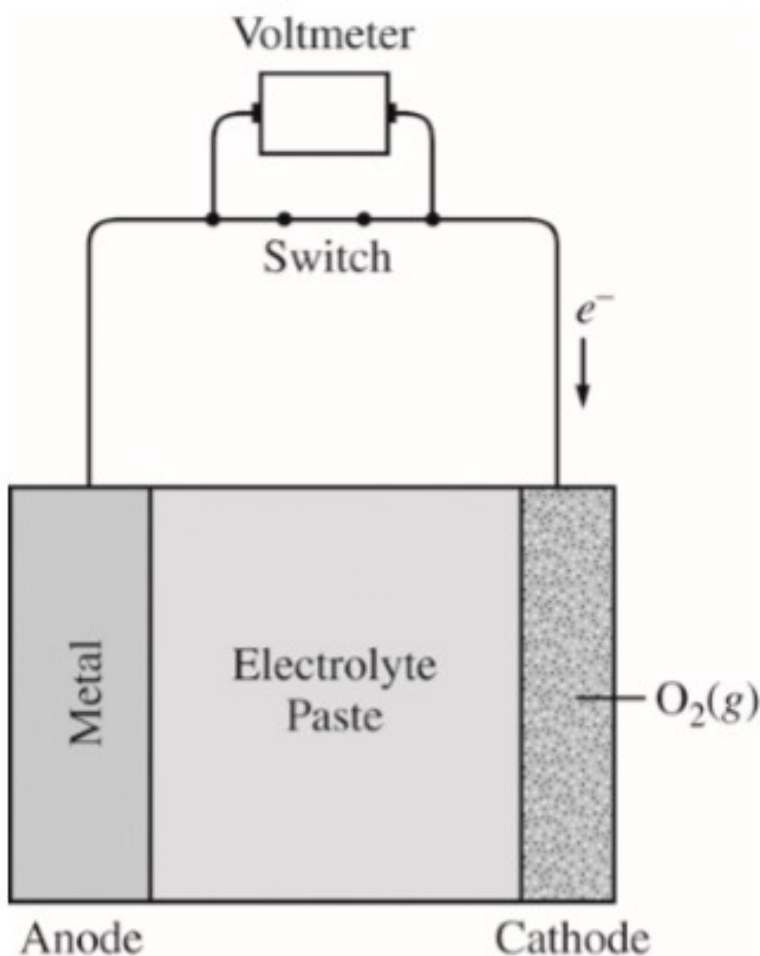


The diagram above shows the experimental setup used to electroplate a small object. A current of 0.80 ampere is supplied to $\text{AgNO}_3(aq)$ for 30.0 minutes. Which of the following equations can be used to calculate the approximate mass of $\text{Ag}(s)$ deposited on the object?

- (A) $\text{Mass} = \left(\frac{0.80 \times 30.0 \times 60}{96,485 \times 107.9} \right) \text{ g}$
- (B) $\text{Mass} = \left(\frac{0.80 \times 30.0 \times 60 \times 96,485}{107.9} \right) \text{ g}$
- (C) $\text{Mass} = \left(\frac{0.80 \times 30.0 \times 60 \times 107.9}{96,485} \right) \text{ g}$ ✓
- (D) $\text{Mass} = \left(\frac{0.80 \times 30.0 \times 107.9}{96,485} \right) \text{ g}$

Chapter 12 Part 2: Electrolysis

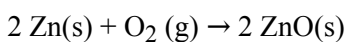
14.



Metal-air cells are a relatively new type of portable energy source consisting of a metal anode, an alkaline electrolyte paste that contains water, and a porous cathode membrane that lets in oxygen from the air. A schematic of the cell is shown above. Reduction potentials for the cathode and three possible metal anodes are given in the table below.

Half Reaction	E at pH 11 and 298 K (V)	(a)
$\text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4 e^- \rightarrow 4 \text{OH}^-(aq)$	+0.34	
$\text{ZnO}(s) + \text{H}_2\text{O}(l) + 2 e^- \rightarrow \text{Zn}(s) + 2 \text{OH}^-(aq)$	-1.31	
$\text{Na}_2\text{O}(s) + \text{H}_2\text{O}(l) + 2 e^- \rightarrow 2 \text{Na}(s) + 2 \text{OH}^-(aq)$	-1.60	
$\text{CaO}(s) + \text{H}_2\text{O}(l) + 2 e^- \rightarrow \text{Ca}(s) + 2 \text{OH}^-(aq)$	-2.78	

Early forms of metal-air cells used zinc as the anode. Zinc oxide is produced as the cell operates according to the overall equation below.



Chapter 12 Part 2: Electrolysis

- i. Using the data in the table above, calculate the cell potential for the zinc-air cell.
- ii. The electrolyte paste contains OH^- ions. On the diagram of the cell above, draw an arrow to indicate the direction of migration of OH^- ions through the electrolyte as the cell operates.
- (b) A fresh zinc-air cell is weighed on an analytical balance before being placed in a hearing aid for use.
- i. As the cell operates, does the mass of the cell increase, decrease, or remain the same?
- ii. Justify your answer to part (b)(i) in terms of the equation for the overall cell reaction.
- (c) The zinc-air cell is taken to the top of a mountain where the air pressure is lower.
- i. Will the cell potential be higher, lower, or the same as the cell potential at the lower elevation?
- ii. Justify your answer to part (c)(i) based on the equation for the overall cell reaction and the information above.
- (d) Metal-air cells need to be lightweight for many applications. In order to transfer more electrons with a smaller mass, Na and Ca are investigated as potential anodes. A 1.0 g anode of which of these metals would transfer more electrons, assuming that the anode is totally consumed during the lifetime of a cell? Justify your answer with calculations.
- (e) The only common oxide of zinc has the formula ZnO .
- i. Write the electron configuration for a Zn atom in the ground state.
- ii. From which sublevel are electrons removed when a Zn atom in the ground state is oxidized?

Part A

- 1 point is earned for the correct cell potential.

$$E_{\text{cell}} = 0.34 \text{ V} - (-1.31 \text{ V}) = 1.65 \text{ V}$$

- 1 point is earned for indicating the movement of OH^- ions from right to left in the cell.

(The arrow should point to the left.)



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for the correct cell potential.

$$E_{\text{cell}} = 0.34 \text{ V} - (-1.31 \text{ V}) = 1.65 \text{ V}$$

Chapter 12 Part 2: Electrolysis

- 1 point is earned for indicating the movement of OH^- ions from right to left in the cell.

(The arrow should point to the left.)

Part B

- 1 point is earned for indicating an increase in cell mass.

The mass increases.

- 1 point is earned for the justification.

Oxygen gas from the air reacts with $\text{Zn}(s)$ in the cell, producing $\text{ZnO}(s)$, which has more mass than the original $\text{Zn}(s)$.



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for indicating an increase in cell mass.

The mass increases.

- 1 point is earned for the justification.

Oxygen gas from the air reacts with $\text{Zn}(s)$ in the cell, producing $\text{ZnO}(s)$, which has more mass than the original $\text{Zn}(s)$.

Part C

- 1 point is earned for indicating a lower cell potential.

The cell potential will be lower.

- 1 point is earned for a justification that relates a lower pressure (or concentration) of $\text{O}_2(g)$ to Q , or a qualitative approach using the Nernst equation.

$\text{O}_2(g)$, a reactant in the cell reaction, will be at a lower partial pressure at the higher elevation; thus the reaction has a greater value of Q (closer to K). Deviations in partial pressure that take the cell closer to equilibrium will decrease the magnitude of the cell potential.



0	1	2
---	---	---

The student response earns two of the following points:

- 1 point is earned for indicating a lower cell potential.

The cell potential will be lower.

Chapter 12 Part 2: Electrolysis

- 1 point is earned for a justification that relates a lower pressure (or concentration) of $O_2(g)$ to Q , or a qualitative approach using the Nernst equation.

$O_2(g)$, a reactant in the cell reaction, will be at a lower partial pressure at the higher elevation; thus the reaction has a greater value of Q (closer to K). Deviations in partial pressure that take the cell closer to equilibrium will decrease the magnitude of the cell potential.

Part D

1 point is earned for the correct calculation of moles for Na and Ca.

$$\text{For Na, } 1.0 \text{ g Na} \times \frac{1.0 \text{ mol Na}}{22.99 \text{ g Na}} \times \frac{1.0 \text{ mol } e^-}{1.0 \text{ mol Na}} = 0.043 \text{ mol } e^-$$

$$\text{For Ca, } 1.0 \text{ g Ca} \times \frac{1.0 \text{ mol Ca}}{40.08 \text{ g Ca}} \times \frac{2.0 \text{ mol } e^-}{1.0 \text{ mol Ca}} = 0.050 \text{ mol } e^-$$

1 point is earned for taking 1 vs. 2 moles of electrons into account and the correct answer.

The cell with the Ca anode would transfer more electrons.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct calculation of moles for Na and Ca.

$$\text{For Na, } 1.0 \text{ g Na} \times \frac{1.0 \text{ mol Na}}{22.99 \text{ g Na}} \times \frac{1.0 \text{ mol } e^-}{1.0 \text{ mol Na}} = 0.043 \text{ mol } e^-$$

$$\text{For Ca, } 1.0 \text{ g Ca} \times \frac{1.0 \text{ mol Ca}}{40.08 \text{ g Ca}} \times \frac{2.0 \text{ mol } e^-}{1.0 \text{ mol Ca}} = 0.050 \text{ mol } e^-$$

1 point is earned for taking 1 vs. 2 moles of electrons into account and the correct answer.

The cell with the Ca anode would transfer more electrons.

Part E

- 1 point is earned for a correct configuration.

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$ or $[Ar]4s^2 3d^{10}$

- 1 point is earned for the correct answer.

4s sublevel



0	1	2
---	---	---

Chapter 12 Part 2: Electrolysis

The student response earns two of the following points:

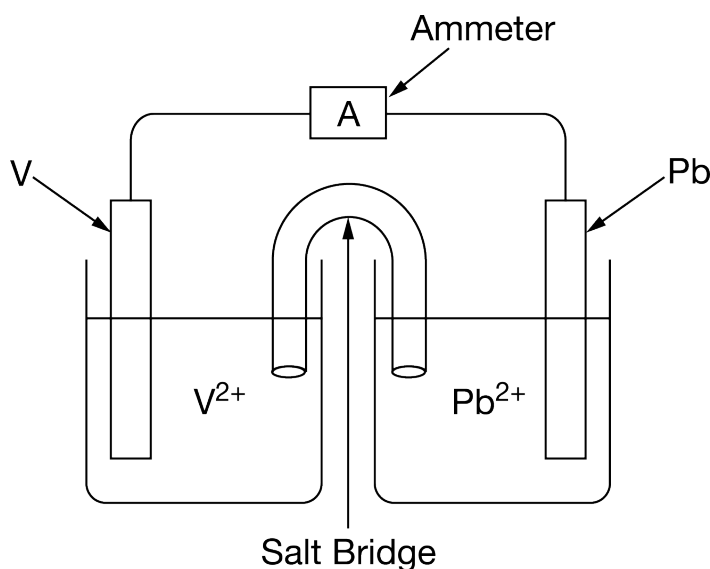
- 1 point is earned for a correct configuration.

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$ or $[\text{Ar}]4s^2 3d^{10}$

- 1 point is earned for the correct answer.

4s sublevel

15.



Reaction 1:	$\text{V}^{2+}(\text{aq}) + 2 e^{-} \rightarrow \text{V}(\text{s})$	$E^{\circ} = -1.18 \text{ V}$
Reaction 2:	$\text{Pb}^{2+}(\text{aq}) + 2 e^{-} \rightarrow \text{Pb}(\text{s})$	$E^{\circ} = -0.13 \text{ V}$
Net Reaction:	$\text{V}(\text{s}) + \text{Pb}^{2+}(\text{aq}) \rightarrow \text{Pb}(\text{s}) + \text{V}^{2+}(\text{aq})$	

The vanadium-lead galvanic cell in the diagram above initially operates under standard conditions according to the net reaction shown in the table below the diagram. The cell operates for 1.0 minute at a constant known current. Which of the following mathematical relationships would be most useful for determining the change in mass of the $\text{V}(\text{s})$ electrode after 1.0 minute of the cell's operation without having to use a balance?

(A) $E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{RT}{nF} \ln Q$.

(B) $I = \frac{q}{t}$

(C) $\Delta G^{\circ} = -nFE^{\circ}$

(D) $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$

